**Department of Information Technology**

Electronic Question Bank

Subject: Computer Architecture and organization Class 3rd sem IT

| SN. | Unit No | | Time Required | Marks allotted | Difficulty Level |
|---|---|---|---|---|---|
| 1. | 1 | Name the functional units of computer and explain how they are interrelated. | 3 | 5 | Easy |
| 2. | | Define the fallowing terms<br>● Random access memory<br>● Memory access time<br>● Memory address<br>● Bus<br>● Operating system. | 3 | 3 | Easy |
| 3. | | List out the operational steps the computer has to perform for program execution . | 7 | 5 | Easy |
| 4. | | Describe the operations related with main memory of computer system.. | 6 | 5 | Easy |
| 5. | | Define one address, two address ,three address instruction. | 7 | 3 | Easy |
| 6. | | What is meant by central processing unit(CPU)? | 5 | 5 | Easy |
| 7. | | Differentiate between little-endian and big endian assignment. | 5 | 5 | Easy. |
| 8. | | Explain Single Bus Organization in computer System. | 5 | 5 | Easy |
| 9. | | Give the connection between processor and memory with neat sketch. | 7 | 5 | Easy |
| 10. | | State the advantages and disadvantages for single bus two bus and three bus organization. | 5 | 5 | Moderate |
| 11. | | Describe the following terms<br>● MAR<br>● MDR<br>● Program counter<br>● Instruction Register<br>● General purpose register. | 5 | 5 | Moderate |
| 12. | | Differentiate between system software and application software. | 6 | 5 | Moderate |
| 13. | | Explain the different addressing modes with example | | | Moderate |
| 14. | | Explain two bus organization with advantages and disadvantages. | 6 | 5 | Moderate |

| | | | | | |
|---|---|---|---|---|---|
| 15. | | Define subroutine. What is the use of link resister explain with example | 5 | 3 | Moderate |
| 16. | | Define stacks. Explain its storage mechanism and use of stack pointer for stack operation. | 7 | 5 | Moderate |
| 17. | | The Memory contains instruction as well as data. How CPU comes to know a particular word of memory contains data or instruction? | 10 | 6 | Difficult |
| 18. | | How to calculate the effective address in each addressing mode. Explain with example. | 10 | 6 | Difficult |
| 19. | | Represent the fallowing instruction in one address two address and three address instruction.<br>X=(a+b)*(c-d)/e. | 10 | 6 | Difficult |
| 20. | | Represent the fallowing instruction in one address two address and three address instruction.<br>X=(a*b)+(c-d)/e | 10 | 5 | Difficult |
| 21. | | Explain the ways in which 32-bit word information can be represented in memory. | 8 | 5 | Difficult |
| 22. | | Write a short note on following<br>1. Distributed system<br>2. Sofware. | 8 | 8 | Difficult |
| 23. | | Both of the following statements cause the value 300 to be stored in location 1000 but at different times<br>ORIGIN 1000<br>DATAWORD 300<br>And<br>Move #300,1000. | 10 | 5 | Very Difficult |
| 24. | | Registers R1 and R2 of a computer contain the decimal values 1200 and 4600 .What is the effective address of the memory operand in each of the following instruction<br>1. Load 20(R1),R5<br>2. Move #3000,R5<br>3. Store R5,30(R1,R2)<br>4. Add -(R2),R5<br>5. Substract (R1)+,R5 | 10 | 5 | Very Difficult |
| 25. | | Consider the following possibilities for saving the return address of a subroutine:<br>(a)In processor register<br>(b)In a memory location associated with the call, so that different location is used when subroutine is called from different places<br>(c) On a stack<br>Which of these possibilities supports subroutine nesting and which supports subroutine recursion. | 8 | 5 | Very Difficult |
| 26. | | List and explain the steps needed to execute the following instructions<br>Add LOCA, R0. | 5 | | Very Difficult |
| 27. | | Represent the fallowing instruction in one address two address and three address instruction. | 8 | 8 | Very Difficult |

| | | | | | |
|---|---|---|---|---|---|
| | | X=(a+b)*(c-d)/(e+f)+(g-h) | | | |
| 28. | | Write a program that can evaluate the expression A*B + C*D | 6 | 5 | Very Difficult |
| 29. | | A memory byte location contains the pattern 00101100.What does this pattern represent when interpreted as a binary number? What does it represent as ASCII code. | 8 | 5 | Very Difficult |
| SN. | Unit No | | Time Required | Marks allotted | Difficulty Level |
| 1. | II | What are the different steps that processor has to perform while executing the instruction. | 3 | 5 | Easy |
| 2. | | List out the limitations of short word length machine. | 5 | 5 | Easy |
| 3. | | What are different high level language considerations. | 7 | 5 | Easy |
| 4. | | Define the following terms.<br>1.Data path<br>2.Fetch phase<br>3.Execute phase. | 3 | 3 | Easy |
| 5. | | What do you mean by bus. State the different types of buses. How the bus is useful for transferring the data. | 7 | 3 | Easy |
| 6. | | Draw the flowchart which will illustrate the basic instruction cycle. | 5 | 5 | Easy |
| 7. | | Write the meaning of the fallowing signals for single bus organisation.<br>MARin, MDRout, Branch>0 ,IRin | 5 | 5 | Easy. |
| 8. | | Write the meaning of following signals for single bus organisation.<br>R1out ,WFMC,CARRY in to ALU. | 5 | 5 | Easy |
| 9. | | Why it is advantageous to use three bus architecture over single bus. | 5 | 5 | Moderate |
| 10. | | Draw and explain the single bus organization. | 5 | 5 | Moderate |
| 11. | | What are the different performance parameter for processor. Explain. | 5 | 5 | Moderate |
| 12. | | Draw and explain the register structure for Motorola 68000 . | 6 | 5 | Moderate |
| 13. | | List out the actions that are required for fallowing.<br>a) Read the word from memory. b) Write the word to memory.<br>c)Move the word from one register to another. | 7 | 5 | Moderate |
| 14. | | Write and explain the actions that are required to perform Move r2,(r1) | 7 | 5 | Moderate |
| 15. | | What is the use of A7 register in Motorola 68000 machine. | 5 | 3 | Moderate |
| 16. | | Draw and explain the status register of 68000. | 5 | 3 | Moderate |

| | | | | | |
|---|---|---|---|---|---|
| 17. | | Write the control sequence for execution of following instruction. SUB (R2),R1 | 7 | 5 | Difficult |
| 18. | | Draw and explain the single bus organization | 7 | 5 | Difficult |
| 19. | | Write the control sequence for execution of following instruction. SUB (R2+),R1 for single bus organization. | | | Difficult |
| 20. | | Draw and explain three bus organization and write control signals for add R3,R4,R1 for three bus structure. | 10 | 6 | Difficult |
| 21. | | How the following data is represented in memory<br>1.Machine Instruction<br>2.Characters<br>3.Numeric data. | 10 | 6 | Difficult |
| 22. | | Draw and explain two bus organization and write control signals for add R3,R4,R1 for two bus structure. | 10 | 6 | Difficult |
| 23. | | How branching is performed. Write the control sequence for execution of unconditional branch instruction. | 7 | 7 | Difficult |
| 24. | | Write and explain the control sequence for the instruction ADD (R4),R3. | 8 | 7 | Difficult |
| 25. | | How branching is performed. Write the control sequence for execution of conditional branch instruction. | 10 | 7 | Very Difficult |
| 26. | | Why is the Wait for memory function completed step needed when reading from or writing to the main memory. | 5 | 5 | Very Difficult |
| 27. | | Explain how the instruction A=B-C gets executed in a 3 bus organization system. | 8 | 5 | Very Difficult |
| 28. | | Write a control steps required for single bus organization for the following instruction<br>Add the (immediate) number NUM to register R1 | 10 | 5 | Very Difficult |
| 29. | | Write a control steps required for single bus organization for the following instruction<br>Add the contents of memory location NUM to register R1. | 10 | 5 | Very Difficult |
| 30. | | Explain input and output gating for registers. | 8 | 5 | Very Difficult |
| 31. | | Show the possible control sequence for implementing the instruction MUL R1,R2. | 5 | | Very Difficult |
| 32. | | For three bus organization show the control steps for branch on negative instruction for processor. | 8 | 8 | Very difficult |
| SN. | Unit No | | Time Required | Marks allotted | Difficulty Level |
| 1. | III | What do mean by emulation . | 3 | 5 | Easy |
| 2. | | Define the following terms.<br>● Micro programmed control.<br>● Control word | 5 | 5 | Easy |

| | | | | | |
|---|---|---|---|---|---|
| | | • Control store | | | |
| 3. | | State the meaning of fallowing terms.<br>• Micro routine<br>• Micro instructions | 7 | 7 | Easy |
| 4. | | What are the different ways to reduce the number of bits in microinstructions. | 3 | 8 | Easy |
| 5. | | What do you mean by bit slicing. | 7 | 3 | Easy |
| 6. | | What are the various factors that reduces the performance of pipelined processor. | 5 | 5 | Easy |
| 7. | | What are the different situations when microprogram counter is not get increment with next address field? | 5 | 5 | Easy. |
| 8. | | Draw and explain the basic organization of a microprogrammed control unit. | 5 | 5 | Moderate |
| 9. | | How to reduce the length of microinstruction explain with example. | 5 | 7 | Moderate |
| 10. | | Explain microinstruction with next address field. | 5 | 7 | Moderate |
| 11. | | Explain prefetching of microinstructions. and state its advantages. | 6 | 5 | Moderate |
| 12. | | With neat diagram explain the basic organization of microprogramed control unit also draw and explain the flowchart for microprogram for add scr, Rdest for index addressing mode. | 7 | 5 | Moderate |
| 13. | | Explain microprogram sequencing. | 7 | 5 | Moderate |
| 14. | | Draw and explain the organization of the control unit to allow conditional branching in micro program. | 5 | 3 | Moderate |
| 15. | | Explain the features of bit slicing. | 5 | 3 | Moderate |
| 16. | | Draw and explain the architecture for hardwired control unit. | | | |
| 17. | | Differentiate between horizontal and vertical organization. | 5 | 3 | Difficult |
| 18. | | Differentiate between hardwired and microprogramed control. | 7 | 5 | Difficult |

| | | | | | |
|---|---|---|---|---|---|
| 19. | | State advantages and disadvantages of microprogrammed control. | | | Difficult |
| 20. | | Why there is need of grouping of control signals explain with example. | 10 | 6 | Difficult |
| 21. | | Write an micro instruction for Add(Rsrc)+,Rdst. | 10 | 6 | Difficult |
| 22. | | For a single bus organization of data paths inside CPU ,write a micro program of micro instructions and draw flowchart of a microprogram for the following instruction.MOV(Rscr)+,Rdst. | 10 | 6 | Difficult |
| 23. | | Write the control sequence for the instruction add[r5],r6 and also write the microinstructions for the same. | 7 | 7 | Difficult |
| 24. | | Draw and explain the organization of the control unit to allow conditional branching in microprogram. | 8 | 7 | Difficult |
| 25. | | Differentiate between hardwired and microprogrammed control. | 8 | 7 | Difficult. |
| 26. | | What do mean by micro program sequencing. | 10 | 7 | Very Difficult |
| 27. | | Explain the different features of macro. | 5 | 5 | Very Difficult |
| 28. | | For a single bus organization of data paths inside CPU ,write a micro program of micro instructions and draw flowchart of a microprogram for the following instruction.add(Rscr)+,Rdst. | 8 | 5 | Very Difficult |
| 29. | | For a single bus organization of data paths inside CPU ,write a micro program of micro instructions and draw flowchart of a microprogram for the following instruction. add(Rscr),Rdst. | 10 | 5 | Very Difficult |
| 30. | | For a single bus organization of data paths inside CPU, write a micro program of micro instructions and draw flowchart of a microprogram for the following instruction. Add Rscr,Rdst. | 10 | 5 | Very Difficult |
| 31. | | Explain the organization of 16 bit sliced processor. | 8 | 5 | Very Difficult |
| 32. | | For a single bus organization of data paths inside CPU, write a micro program of micro instructions and draw flowchart of a microprogram for the following instruction. sub Rscr,Rdst. | 5 | 7 | Very Difficult |
| 33. | | For a single bus organization of data paths inside CPU ,write a micro program of micro instructions and draw flowchart of a microprogram for the following instruction.add -(Rscr),Rdst. | 8 | 8 | Very Difficult |
| SN. | Unit No | | Time Required | Marks allotted | Difficulty Level |
| 1. | IV | What are the different formats for number representation? Give example of each. | 3 | 3 | Easy |
| 2. | | Write binary rules for arithmetic addition and subtraction. | 5 | 3 | Easy |

| | | | | | |
|---|---|---|---|---|---|
| 3. | | Give the rules for Multiplication, division, addition, subtraction of floating point numbers. | 7 | 5 | Easy |
| 4. | | Write a logic truth table for sum and carry -out functions for adding equally weighted bits xi and yi in two numbers X and Y and implement it with 3 input X-OR gate. | 3 | 5 | Easy |
| 5. | | Give comparison between restoring and non-restoring division in terms of speed, Time .Which technique is best justify your answer. | 7 | 3 | Easy |
| 6. | | Draw the table for multiplicand selection decisions in bit pair recording method and for booths multiplication method. | 5 | 4 | Easy |
| 7. | | Consider the binary numbers in following addition and subtraction problems to be signed,6 bit values in 2's compliment representation. Perform the operations indicated, specify whether arithmetic overflow occurs.<br>010110 110111 011010 100001<br>+ 001001 +100101 - 100010 -011101 | 5 | 5 | Easy. |
| 8. | | Represent the fallowing numbers in 1's compliment 2's compliment and sign and magnitude format.<br>i) 110111 ii) 010110 | 5 | 5 | Easy |
| 9. | | Define guard bits. What are the different methods of truncation? Explain each with example. | 5 | 7 | Moderate |
| 10. | | Draw and explain the logic design for division. Write an algorithm for restoring division technique and solve 18/4. | 5 | 8 | Moderate |
| 11. | | State the advantages of bit pair recording method over booths algorithm and Solve the following<br>i)-12*6(booths algorithm)<br>ii)-14*7(bit pair recording method) | 5 | 7 | Moderate |
| 12. | | Explain the logic design for fast address. | 6 | 5 | Moderate |
| 13. | | Solve the following using sequential multiplier. 1101*1010. | 5 | 5 | Moderate |
| 14. | | Solve the following using sequential multiplier. 0101*1011. | 5 | 5 | Moderate |
| 15. | | Illustrate multiplication of signed 2's compliment numbers 01101 and 11010 using bit pairing of the multipliers. | 7 | 7 | Moderate |
| 16. | | Draw and explain the logical circuit for array multiplication. | 5 | 5 | Moderate |

| | | | | | |
|---|---|---|---|---|---|
| 17. | | Perform 1100 by 0011 using non-restoring algorithm. | 5 | 7 | Moderate |
| 18. | | Multiply each of the following pairs of signed 2's compliment numbers using booth algorithm assume that A is multiplicand and B is multiplier.<br>i) A=010111 B=110110<br>ii) A=110011 B=101100 | 7 | 7 | Moderate |
| 19. | | Multiply each of the following pairs of signed 2's compliment numbers using booth algorithm assume that A is multiplicand and B is multiplier.<br>i) A=110101 B=011011<br>ii) A=001111 B=001111 | 7 | 7 | Moderate |
| 20. | | Multiply each of the following pairs of signed 2's compliment numbers using bit pair recording method assume that A is multiplicand and B is multiplier.<br>i) A=010111 B=110110<br>ii) A=110011 B=101100 | 7 | 7 | Moderate |
| 21. | | Multiply each of the following pairs of signed 2's compliment numbers using bit pair recording method assume that A is multiplicand and B is multiplier.<br>i) A=110101 B=011011<br>ii) A=001111 B=001111 | 7 | 7 | Moderate |
| 22. | | Draw and explain the logic design for division. Write an algorithm for non-restoring division technique and solve 13/4. | 10 | 6 | Difficult |
| 23. | | Draw and explain the logic design for division. Write an algorithm for restoring division technique and solve 13/4. | 10 | 6 | Difficult |
| 24. | | List the advantages of fast multiplication over booths algorithm and solve (+13) *(-6) using fast multiplication. | 10 | 6 | Difficult |
| 25. | | Write an booths algorithm and solve (-13) *(11) using booths algorithm. | 7 | 7 | Difficult |
| 26. | | Explain basic principle of Booth's algorithm and using bit pair recoding method, multiply the following pair of signed 2's complement numbers: -<br>A=010111 B=110110 | 8 | 7 | Difficult |
| 27. | | 11.Explain IEEE floating point formats and represent following numbers in them: -<br>a. 0.1010010<br>b. (1.00101) 10 | 10 | 7 | Difficult |
| 28. | | Explain basic principle of Booth's algorithm and using bit pair recoding method, multiply the following pair of signed 2's complement numbers: -<br>A=010111 B=110110 Where A is multiplicand and B is multiplier. | 5 | 5 | Difficult |

| | | | | | |
|---|---|---|---|---|---|
| 29. | | Explain with the help of block diagram working of a floating-point addition/subtraction unit. | 8 | 5 | Difficult |
| 30. | | Explain booths algorithm to multiply the following pair of signed two's compliment numbers A=110011 multiplicand B=101100 multiplier Also implement the above using bit-pair recording and explain how it achieves faster multiplication. | 8 | 7 | Difficult |
| 31. | | Draw and explain logic design for binary multiplication. And solve 15*11 using booths algorithm. | 8 | 7 | Difficult |
| 32. | | Explain technique "carry-save addition of Summands" for making multiplication process faster. | 8 | 7 | Difficult |
| 33. | | Indicate how to modify the circuit diagram of register configuration of binary multiplier to implement multiplication of signed 2's compliment ,n bit numbers using booth algorithm ,by clearly specifying inputs and outputs for the control sequencer and any other changes needed around the adder and A register. | 10 | 7 | Very Difficult |
| 34. | | If the product of two n bit signed numbers in 2's compliment representation can be represented in n bits, the manual multiplication algorithm can be used directly, treating the sign bits same as other bits. Implement this on each of the following pairs of 4-bit signed numbers.<br>i) Multiplicand =1110 Multiplier=1101<br>ii) Multiplicand =0010 Multiplier=1110<br><br>Why it works correctly. | 10 | 7 | Very Difficult |
| 35. | | A half adder is a combinational logic circuit that has two inputs x and y and two outputs s and c that are sum and carry-out respectively ,resulting from binary addition od x and y<br>i) Design a half adder as a two level AND-OR circuit<br>ii) Show how to implement full adder by using two half adders and external logic gates as necessary. | 8 | 7 | Very Difficult |
| 36. | | Using manual method perform the operations A * B on 5 bit unsigned numbers A=10101 B=00101. How this multiplication would be performed by the circuit arrangement of binary multiplication . | 5 | | Very Difficult |
| 37. | | Using manual method perform the operations A / B on 5 bit unsigned numbers A=10101 B=00101. How this division would be performed by the circuit arrangement of binary Division. | 8 | 8 | Very Difficult |
| 38. | | Consider the 16-bit floating point number in format with 6 bit exponent and 9 bit normalized fractional | 6 | 5 | Very Difficult |

<table>
<tr><td></td><td></td><td>mantissa.The base of the scale factore is 2 and exponent is represented in excess-31 format<br>(a) Add the numbers A and B formatted as follows:<br>A=<br><table><tr><td>0</td><td>100001</td><td>111111110</td></tr></table><br>B=<br><table><tr><td>0</td><td>011111</td><td>001010101</td></tr></table><br>Give the answer in normalized form. Use rounding when producing normalized the final normalized 9 bit mantissa.</td><td></td><td></td><td></td></tr>
<tr><td>39.</td><td></td><td>Consider that floating point numbers are represented in a 12 bit format .The scale factor has an implied base 2 and a 5-bit excess-15 exponent with the two end values of 0 and 31 used to signify exact 0 and infinity. The 6 bit mantissa is normalized as in IEEE format with an implied 1 to the left of binary point .The 6 bit mantissa is normalized in IEEE format with implied 1 to the left of the binary point.<br><table><tr><td>1 bit sign bit</td><td>5 bit excess-15 exponent</td><td>6 bit fractional mantissa</td></tr></table><br>i) Represent +1.7 ,-0.012,+19 and 1/8 in this format.<br>ii) What are the smallest and largest number represented in this format.</td><td>8</td><td>7</td><td>Very Difficult</td></tr>
<tr><td>40.</td><td></td><td>Perform Add ,Subtract, Multiply and divide operations on the operands.<br>A=<br><table><tr><td>0</td><td>10001</td><td>011011</td></tr></table><br>B=<br><table><tr><td>1</td><td>01111</td><td>101010</td></tr></table></td><td>10</td><td>6</td><td>Very difficult</td></tr>
<tr><td>SN.</td><td>Unit No</td><td></td><td>Time Required</td><td>Marks allotted</td><td>Difficulty Level</td></tr>
<tr><td>1.</td><td>V</td><td>Define the following terms.<br>● Memory access Time<br>● Memory cycle Time<br>● Random Access Memory<br>● Cache Memory</td><td>5</td><td>5</td><td>Easy</td></tr>
</table>

| | | | | | |
|---|---|---|---|---|---|
| | | • Virtual Memory | | | |
| 2. | | Differentiate between static and dynamic memory's. | 5 | 3 | Easy |
| 3. | | How read and write operations are performed on SRAM cell. | 7 | 5 | Easy |
| 4. | | Selection of the RAM chip for particular application is depend on which factors. | 5 | 4 | Easy |
| 5. | | Explain the following terms related with memory Bandwidth and latency time | 7 | 3 | Easy |
| 6. | | Draw and explain single transistor dynamic memory cell. | 5 | 4 | Easy |
| 7. | | Draw and explain a storage cell for a bipolar read-only memory. | 5 | 5 | Easy. |
| 8. | | List the advantages and disadvantages of different mapping techniques for cache memory. | 5 | 5 | Easy |
| 9. | | Define the following terms<br>• physical address.<br>• logical address.<br>• Word line<br>• Bit line | 3 | 5 | Easy |
| 10. | | Why there is a need of cache memory. | 5 | 3 | Easy |
| 11. | | Draw and explain the internal organization of memory chip. | 5 | 5 | Moderate |
| 12. | | Why there is need for memory management. | 5 | 5 | Moderate |
| 13. | | How memory loading time of block is reduced if module structure of memory is interleaved explain with example. | 5 | 5 | Moderate |
| 14. | | What do you mean by locality of reference? Explain with example. | 6 | 5 | Moderate |
| 15. | | What are the different memory units that are employed effectively in computer system? | 7 | 5 | Moderate |
| 16. | | With respect to cache memory explain the following:<br>a) Locality of reference<br>b) Dirty bit<br>c) Replacement algorithms<br>d) Write-though and write-back policy | 7 | 5 | Moderate |
| 17. | | Explain concept of pipelining. Give an example of 4 stage pipelining. | 5 | 5 | Moderate |
| 18. | | Draw and explain virtual memory organization. | | | Moderate |
| 19. | | What is TLB(translation Look-aside buffer)and what is function of it. | 7 | 5 | Moderate |
| 20. | | Draw and explain the computer memory hierarchy .Compare all memory in terms of size cost and speed. | 7 | 7 | Difficult |

| | | | | | |
|---|---|---|---|---|---|
| 21. | | Explain with block diagram the address translation scheme in virtual machine. | 10 | 7 | Difficult |
| 22. | | A block set-associative cache consists of a total of 64 blocks divided into 4 blocks sets. The main memory contains 4096 blocks, each consisting of 128 words.<br>i)How many bits are there in main memory address?<br>ii)How many bits are there in each of the TAG, SET and WORD fields? | 10 | 6 | Difficult |
| 23. | | What the different mapping techniques of cache memory. Explain with example. | 7 | 6 | Difficult |
| 24. | | What are the differences among direct mapping, associative mapping and set associative mapping. | 8 | 7 | Difficult |
| 25. | | Explain multiple module memories with example. | 10 | 5 | Difficult |
| 26. | | Page replacement strategies are needed only in associative and set associative caches .justify this statement with proper example. | 5 | 5 | Difficult |
| 27. | | An 8 way set associative cache is used in computer in which the real memory size is $2^{32}$<br><br>Bytes .The line size is 16 bytes and there are $2^{10}$ lines per set. Calculate cache size and tag length. | 8 | 5 | Difficult |
| 28. | | Design 8K * 8bits RAM system, using a 1K*4 bits RAM ICs and appropriate decoders. | 10 | 5 | Very Difficult |
| 29. | | Draw block diagram for designing 4 M*8 memory using 256 K *1 memory chips. | 10 | 5 | Very Difficult |
| 30. | | Explain Least recently used(LRU)replacement algorithm with example. | 8 | 5 | Very Difficult |
| 31. | | Given page reference string:with 4 pages per frame<br>1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6<br>Compare the number of page faults for<br>FIFO and Optimal page replacement algorithm. | 5 | 7 | Very Difficult |
| 32. | | Given page reference string: with 4 pages per frame.<br>1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6<br>Compare the number of page faults for LRU,<br>FIFO page replacement algorithm. | 8 | 8 | Very Difficult |
| 33. | | Discuss in qualitative terms the impact of the following design decision on cache performance :<br>(a) Selection of cache size that is too small.<br>(b) Selection of cache block size that is too big.<br>(c) Selection of associativity level k that is too small. | 6 | 5 | Very Difficult |
| 34. | | The main memory of a computer is organized as 64 blocks with block size of eight words. The cache has eight block frames. In the following parts, show the mappings from the numbered blocks in main | 12 | 8 | Very Difficult |

| | | | | | |
|---|---|---|---|---|---|
| | | memory to block frames in the cache. Draw all lines showing the mapping as clear as possible.<br>i) Show the direct mapping and address bits that identify tag field, the block number and the word number.<br>ii) Show full associative mapping and address bits that identify the tag field and word number.<br>iii) Show two way set associative mapping and address bits that identify the tag field, set number and word number. | | | |
| 35. | | Which of the L1 or L2 cache is faster? Which is smaller? Why it is smaller. | 6 | 4 | Very difficult |
| Sr.No | **Unit No** | | Time Required | Marks allotted | Difficulty Level |
| 1. | VI | Define the fallowing terms<br>● Memory mapped I/O<br>● I/O mapped I/O<br>● Interrupts | 5 | 5 | Easy |
| 2. | | What is non -makeable interrupts? What is the action performed on the receipt on NMI? | 5 | 3 | Easy |
| 3. | | Specify the different type of DMA transfer techniques? | 7 | 5 | Easy |
| 4. | | What do you mean by bus arbitration also state the approaches for bus arbitration? | 5 | 5 | Easy |
| 5. | | Define pipelining. Give any real time example for pipelining | 7 | 3 | Easy |
| 6. | | What do mean by Interrupt Service Routine. Explain with example. | 5 | 4 | Easy |
| 7. | | List the different ways of handling data transfer to and from processor with its advantages and disadvantages. | 5 | 5 | Easy. |
| 8. | | Define the fallowing terms<br>● Interrupt latency<br>● Interrupt vector | 5 | 5 | Easy |
| 9. | | What is DMA? And state the need of DMA. | 5 | 5 | Moderate |
| 10. | | State the disadvantages of programed I/O | 5 | 5 | Moderate |
| 11. | | Explain how the keyboard is attached with computer system. | 5 | 5 | Moderate |
| 12. | | Why programmed controlled I/O is unsuitable for high speed data transfer. | 6 | 5 | Moderate |
| 13. | | What are the necessary operations needed to start an I/O operation using DMA. | 7 | 5 | Moderate |

| | | | | | |
|---|---|---|---|---|---|
| 14. | | Differentiate between<br>● Hardware and software interrupts.<br>● Vectored and non-vectored interrupts<br>● Maskable and non-maskable interrupts. | 7 | 6 | Moderate |
| 15. | | Why does DMA have priority over CPU when both request a memory transfer and state why are read and write control lines of DMA controller are bi-directional. | 5 | 7 | Moderate |
| 16. | | What different type of information is send by processor to DMA controller when it wants to read and write block of data. | | | Moderate |
| 17. | | Explain the basic functions of linker and loader | 5 | 5 | Moderate |
| 18. | | How to enable and disable the interrupts. | 7 | 5 | Difficult |
| 19. | | Compare the characteristics of RISC and CISC processors | 7 | 7 | Difficult |
| 20. | | Compare programmed I/O and interrupt-initiated I/O on fallowing attributes<br>Concept, Processor involvement, Implementation, stack initialization, Performance. | 10 | 6 | Difficult |
| 21. | | Explain with diagram how the common interrupt request line is used to serve n devices. | 10 | 6 | Difficult |
| 22. | | Discuss the complete working of DMA controller. | 7 | 6 | Difficult |
| 23. | | What are the different regist`ers used in data transfer operations ?explain each. | 8 | 5 | Difficult |
| 24. | | Explain how the open drain bus is used to implement a common interrupt request line. | 10 | 5 | Difficult |
| 25. | | After enabling the interrupts what are the different steps fallowed by the processor. | 5 | 5 | Difficult |
| 26. | | Differentiate between synchronous and asynchronous bus. | 8 | 5 | Very Difficult |
| 27. | | Explain daisy chain priority interrupts with neat diagram. | 10 | 7 | Very Difficult |
| 28. | | Explain parallel priority interrupts with neat diagram. | 10 | 5 | Very Difficult |
| 29. | | What do mean by polling .Explain | 8 | 5 | Very Difficult |
| 30. | | Explain the centralized arbitration of bus system. | 5 | 7 | Very Difficult |
| 31. | | Explain the Distributed arbitration of bus system. | 8 | 8 | Very Difficult |
| 32. | | Consider the fallowing situation. A number of devices are attached to processor and are capable of initiating the interrupt request ,these devices are operationally independent .<br>Ans the fallowing<br>How processor recognize the device requesting an interrupt? | 6 | 8 | Very Difficult |

| | | | | | |
|---|---|---|---|---|---|
| | | How should two or more simultaneous request be handled. | | | |
| 33. | | Draw and explain the connection between printer to processor. | 12 | 8 | Very Difficult |
| 33. | | Why assembler is design in two passes. What is purpose of pass-I and pass-II of assembler. Draw and explain the flowchart for pass-II. | 8 | 5 | Difficult |
| | | | | | |
| | | | | | |

Submitted By

Mrs.Swati G.Kale

Dept. of IT

**UNIT-I**

PART-A

What are active attacks? How is it classified?
Differentiate cryptography and Stegnography with an example.
Mention the mechanism implemented by the confidentiality and traffic flow confidentiality services.
What is meant by services?
Define the following terms (a) Stream cipher (b) Block cipher.
Differentiate between mono alphabetic cipher and a poly alphabetic cipher.
What is meant by transpositional techniques?
What are the types of security threats?

Define decryption.
Define Ceaser cipher.
What is private key?
Define cipher text.
List and briefly define categories of security mechanism.
Differentiate between active attack and passive attack.
Write a note on playfair cipher.
Why it is important to study the Feistal cipher?
Distinguish between diffusion and confusion.

**PART-B**

1. What is meant by cryptanalysis? What are the types of cryptanalytic attacks and the amount of information known to cryptanalyst? What are the features of these attacks?
2. Discuss in detail about the classical encryption techniques.
3. Explain about model for network security.
4. Explain about encryption algorithms.
5. Compare and contrast substitution and transposition ciphers. Explain one method in each category.
6. How simplified DES scheme is used for encryption/decryption? Discuss.
7. Why DES algorithm is implemented? Explain in detail about the function about DES and compare it with simplified DES.

**UNIT-II**

PART-A

1.Specify the application of public key cryptography.
2.What is the purpose of S-boxes in DES?
3.How is S-box constructed?
4.Distinguish between stream and block ciphers.
5. List three classes of polynomial arithmetic.
6. Define plain text.
7. Differentiate between plain text and cipher text.
8. List out the types of cryptographic functions.
9. What is Elliptic curve cryptography?
10. Differentiate between public key encryption versus conventional encryption.
11. Define authentication.
12. What is modular arithmetic?
13. Define Euler's theorem.
14. Mention the four possible approaches to attack the RSA algorithm.
15. Write the purpose of Diffie –Hellman Key Exchange.
16. What is a public key certificate?

PART-B

1. How is RSA algorithm used in encryption/Decryption? Discuss with example.
2. Explain in detail about distribution of public key and secret key.
3. Explain the concept of multiple encryptions with an example.
4. What are the requirements for public key cryptography? What are the issues in public key cryptanalysis?
5. Explain in brief about Elliptic Curve cryptography.
6. Describe about groups, rings, modular arithmetic and finite field.

**UNIT-III**

PART-A

1. What are the requirements of a hash functions.
2. Illustrate denial of service. Give an example.
3. What are the properties of a hash function used in message authentication.
4. Compare SHA-1 and MD5 algorithm.
5. What is meant by message digest?
6. List out the design objectives for HMAC.
7. What are the three classes of message authentication functions?
8. What is the need for mutual authentication protocol?
9. What are the requirements of digital signature?
10. List out the properties of digital signature.
11. What are the key requirements of message digests?
12. Draw the diagram of TCP segment.
13. What is meant by brute force attack?
14. Define birthday attack.
15. What is web security?
16. Define CMAC.

PART-B

1. Explain simple hash function with example?
2. How hash functions are used in message authentication? Discus in detail about simple hash functions and block chaining techniques.
3. Explain the MD-5 and SHA-1 hashing functions with an example.
4. Explain the digital signature algorithm with example and its application.
5. What are the different types of arbitrated digital signatures? Explain their operation and other features.
6. Describe HMAC algorithm.
7.Explian in detail about authentication function.

**UNIT-IV**

PART-A

1. List out the reasons for using PGP.
2. Why e-mail compatibility function in PGP needed?
3. What is e-mail?
4. Mention any two applications of IP security.
5. Give the benefits of IP security.
6. Specify the IP security services.
7. What are the two aspects of key management in IPSec?
8. Write the functions of S/MIME.
9. What is IP spoofing?
10. State the properties of digital signature.
11. What is meant by session key component?
12. Write about the benefits of IP security.
13. Define SSL connection.
14. What is Kerberos?
15. Write the four notation for PGP.
16. What are the uses of KDC?
17. List out the header fields in MIME.
18. What are the address formats used in e-mail.?

PART-B

1.Briefly discuss about SSL architecture and SSL record protocol.
2.Explain the transmission and reception of PGP messages, with relevant flowcharts, in detail.
3.Explain about ESP format and transport modes of ESP.
4.Explain the implementation of IPSec in detail.
5.How is secure electronic transaction implemented? Explain.
6.Explain in detail about key management in IP Security.
7.Describe about Secure Electronic Transaction.

**UNIT-V**

PART-A

1.What are the design goals of firewalls?
2.List out the limitations of firewall.
3.List out the current standards.
4.Write replay attack with an example.
5.What is trusted system? How is it useful?
6.What is the use of trusted system.
7.What is meant by tiny fragment attack? How can it be defeated?
8.Mention the principle of a packet filtering type of firewall.

9.List out the configuration of firewall.
10.What is an encrypted tunnel?
11.List and briefly define three classes of intruder.
12.Draw the diagram of agent architecture.
13.What are the two common techniques used to protect a password file?
14.List out the four basic techniques for password selection strategies.
15.What is a honey pot?

PART-B

1. What are the characteristics and capabilities of firewalls? Explain the types of firewalls.
2. Explain in detail about internet infrastructure security.
3. Discuss in detail about the concepts of trusted systems and data access control.
4. Explain in detail about prevention of Denial of Service attacks.
5. Illustrate, with necessary diagrams, three common firewall configurations.